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Questions

Q1.

Plastics are carbon-based polymers made from repeating monomers.

Many plastics do not break down naturally and their disposal is causing a lot of damage to the environment.

(a) Plastics have been around for about 70 years.

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Scientists think that microorganisms could evolve the ability to break down plastics to obtain nutrients.

(i) Explain why scientists think that plastics could be broken down by microorganisms.

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(ii) Explain how microorganisms could evolve to break down plastics.

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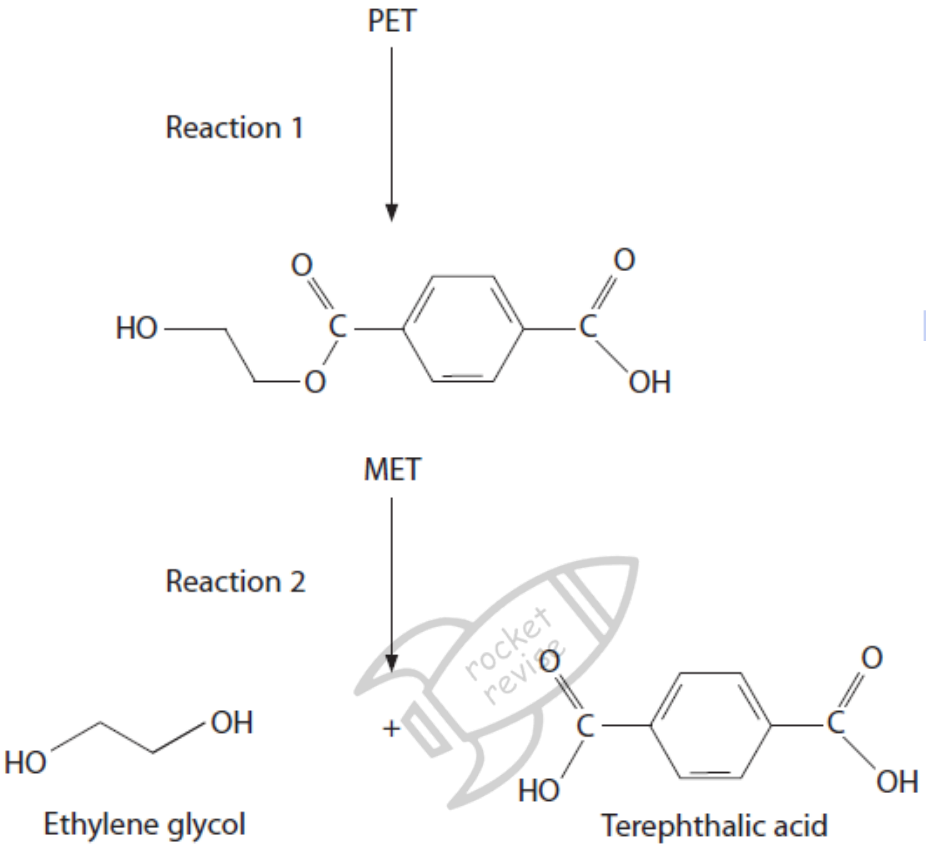
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(b) Some bacteria have been discovered that can break down one type of plastic, the polymer polyethylene terephthalate (PET).

This polymer is made from the monomer ethylene terephthalate (MET).

The diagram shows how PET can be broken down.



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Which row of the table describes reaction 1 and reaction 2?

(1)

	Reaction 1	Reaction 2
☒ **A**	condensation	condensation
☒ **B**	condensation	hydrolysis
☒ **C**	hydrolysis	condensation
☒ **D**	hydrolysis	hydrolysis

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(Total for question = 6 marks)

Q2.

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Malaysia is covered by large areas of rainforest. These rainforests have one of the highest biodiversities in the world.

However, the rate of loss of rainforest in Malaysia is one of the highest in the world.

(a) (i) From 2000 to 2012, Malaysia lost 14.4% of its rainforest. This is equivalent to 47 278 km^2.

Calculate the area of Malaysia covered by rainforest in 2000.

Give your answer in standard form.

(3)

Answer ..... km^2

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(ii) A decrease in the area covered by rainforest results in both a decrease in biodiversity and a decrease in populations.

Explain why a decrease in the area covered by rainforest reduces both biodiversity and populations within the rainforest.

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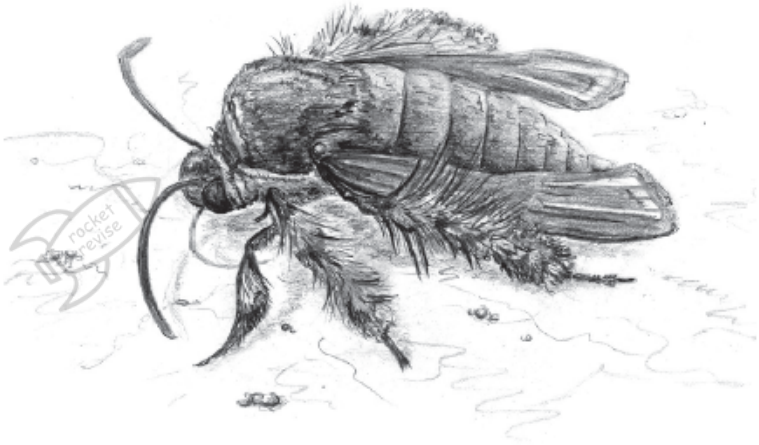
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(b) The diagram shows an Oriental blue clearwing, a species of moth found in the rainforests of Malaysia.

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Magnification ×5

The Oriental blue clearwing has blue bands on its body. It resembles a bee by having transparent wings and fur on its body. It drinks in puddles of water where a number of species of bee also drink.

(i) Explain which examples of adaptation are illustrated by this moth.

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(ii) Suggest **two** advantages to the Oriental blue clearwing of resembling bees.

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(iii) This moth was seen and a specimen collected in 1887. It was not seen again in the rainforests of Malaysia until 2017.

The DNA of these recently rediscovered Oriental blue clearwings was compared with the DNA of the moth collected in 1887.

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Describe how this comparison could be made.

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(Total for question = 12 marks)

Q3.

Body temperature and the degree of muscle contraction can be used to determine the time since death of a person.

The table shows how body temperature and body stiffness, due to muscle contraction, change with time since death.

Time since death / hours	Body temperature	Body stiffness
< 3	warm	not stiff
3 to 8	warm	stiff
8 to 36	cold	stiff
> 36	cold	not stiff

(a) State how the temperature of a dead body should be measured.

(1)

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(b) (i) Body temperature can be used to estimate the time since death using the following information:

- loss of 0.78°C per hour for the first 12 hours after death
- after 12 hours, loss of 0.4°C per hour.

Estimate the time since death of a person whose body temperature had fallen 11.5°C.

Give your answer to the nearest hour.

(2)

Answer ..... hours

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(ii) Explain why this estimate would be different if the body had been left in a colder place.

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(c) Explain why using body stiffness only, as shown in the table, is insufficient to estimate the time since death accurately.

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(Total for question = 8 marks)

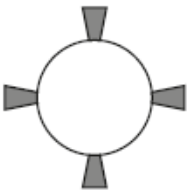
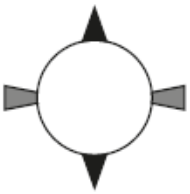
Q4.

Humans can have one of four different blood types: A, B, AB or O.

Blood type is determined by antigens present on the membranes of red blood cells.

The table shows which antigens are present in each blood type.

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Blood type	Antigens on the membrane of red blood cells	Diagram of one red blood cell
A	antigen A only	
B	antigen B only	
AB	both antigens A and B	
O	neither antigen A nor B	

Blood transfusions are used in the treatment of some diseases.

A blood transfusion involves taking blood from a healthy person and putting it into the person needing the treatment.

(a) (i) Which blood type can be used for all blood transfusions?

(1)

☒ **A** blood type A

☒ **B** blood type B

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☒ **C** blood type AB

☒ **D** blood type O

(ii) Explain what will happen if a person with blood type A is given a transfusion of blood type AB.

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(b) Bacteria live in our intestines (gut flora).

The membranes of cells lining our intestines have molecules similar to the antigens on red blood cells.

(i) Explain why bacteria living in our intestines are important.

(2)

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(ii) Both the molecules on the membranes of the cells lining our intestines and the antigens on red blood cells have
sugars attached to protein molecules.

Bacteria living in our intestines secrete an enzyme that separates the sugars from the protein molecules.

Suggest why bacteria living in our intestines secrete this enzyme.

(2)

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(iii) Explain why this enzyme may be useful in blood transfusions.

(2)

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(Total for question = 11 marks)

Q5.

Climate change is dependent on the balance of carbon released into the atmosphere and carbon removed from the atmosphere.

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Some ecosystems are better than others at storing carbon in their soil, releasing less back into the atmosphere.

Conserving these ecosystems could be an important way of reducing climate change.

One study investigated the extent of decomposition in different ecosystems.

This study used tea in teabags as the source of organic matter for decomposition.

The photograph shows a teabag.



(Source: © Hugh Threlfall / Alamy Stock Photo)

(a) The first part of the study was carried out in a laboratory.

Two types of tea were used, green tea and rooibos tea.

A number of unused teabags containing each type of tea were buried in soil at two different temperatures, 15 °C and 25 °C.

The mass of tea in each teabag had been determined before they were buried.

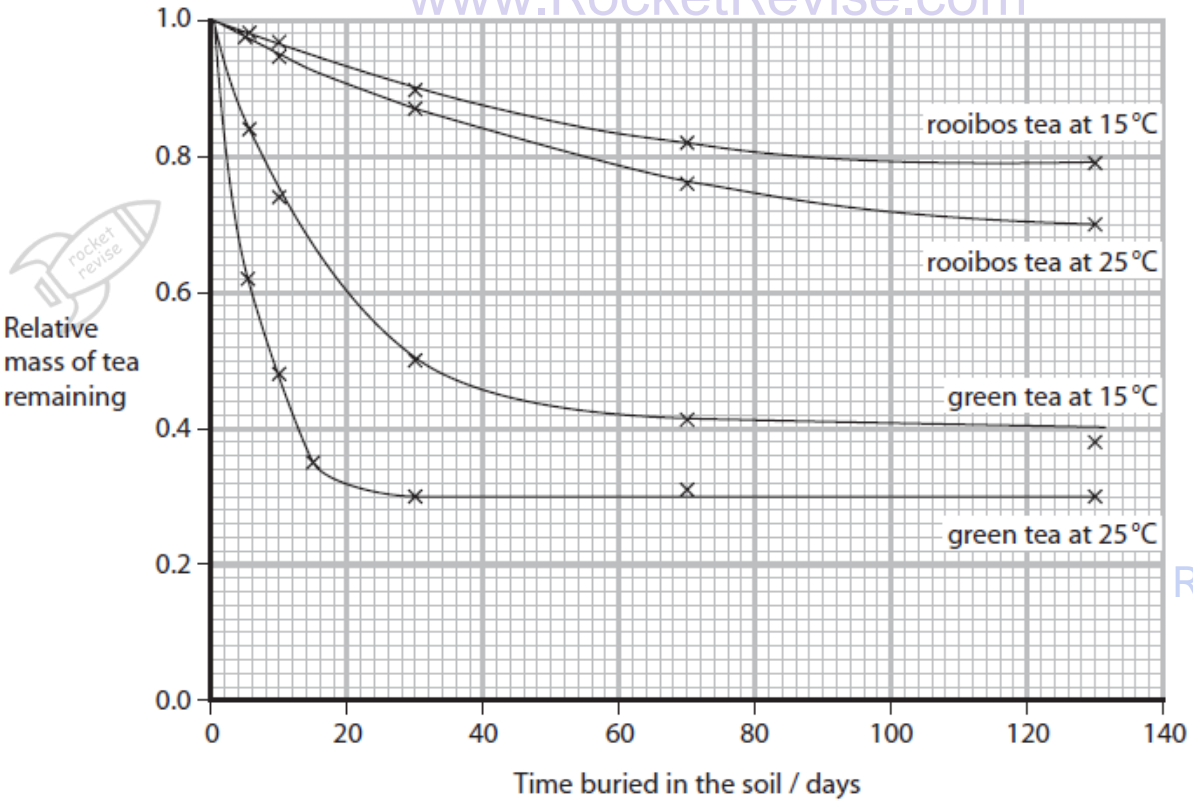
At regular intervals teabags were dug up, dried, and the remaining tea reweighed.

The graph shows the results of this study.

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(i) The teabags had to be dried before weighing to remove water.

One teabag contained 28 g of tea when it was buried. When it was dug up it had a wet mass of 42 g. The water content of this teabag was calculated to be 50 %.

Calculate the mass of organic matter lost during this study.

(1)

Answer ..... g

(ii) Calculate the rate at which the relative mass of the green tea decreases in the teabags buried at 15 °C at day 30.

(2)

Answer .....

(iii) Explain the difference in the decrease in relative mass of green tea in teabags buried at 15 °C and 25 °C.

Use the information in the graph to support your answer.

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(iv) Suggest why the rate of decomposition of green tea was different from that of rooibos tea.

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(b) In the second part of this study, several teabags were buried in different ecosystems.

The teabags were left buried for three months and then dug up.

A number of measurements were taken from the tea in these teabags and the mean decomposition rate and the mean stabilisation factor (S) calculated.

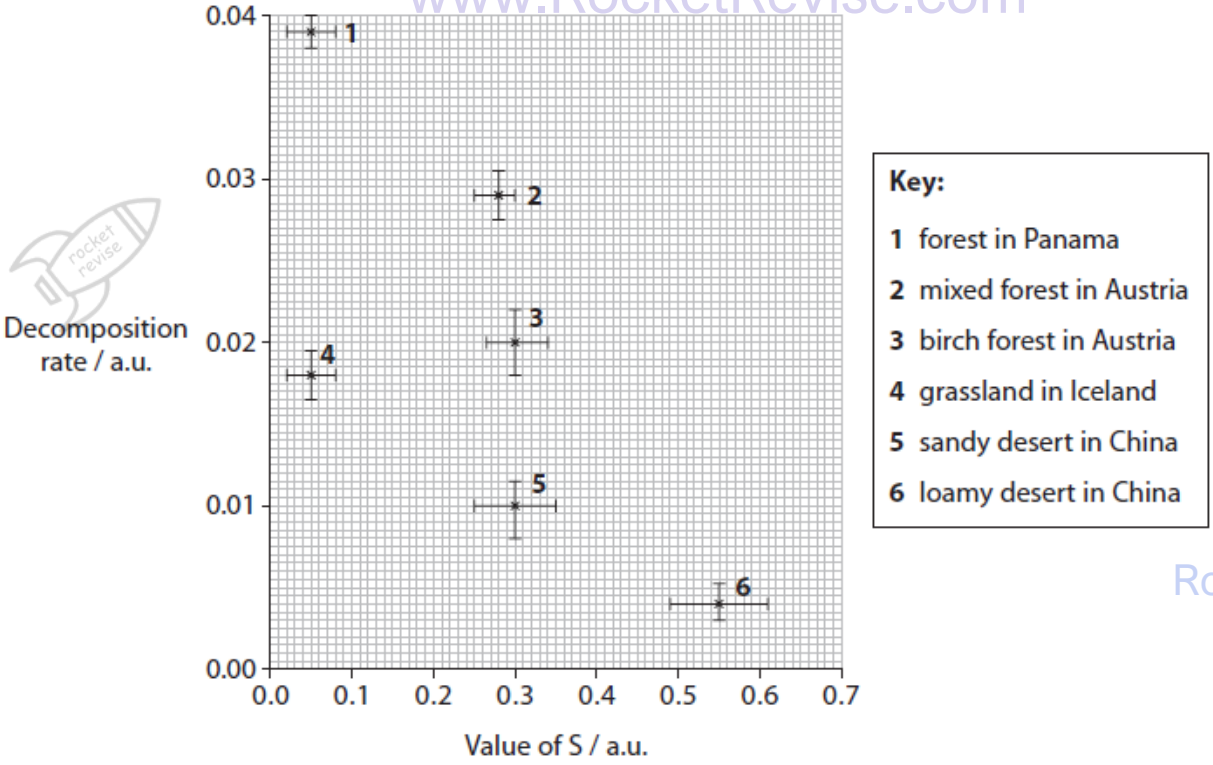
The higher the value of S, the more carbon is stored in the ecosystem.

The graph shows some of the results from this study.

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(i) Suggest why each point on the graph has both a horizontal and a vertical error bar plotted with it.

(1)

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(ii) Explain which ecosystems should be conserved to have the greatest impact on climate change.

Use the information in the graph to support your answer.

(3)

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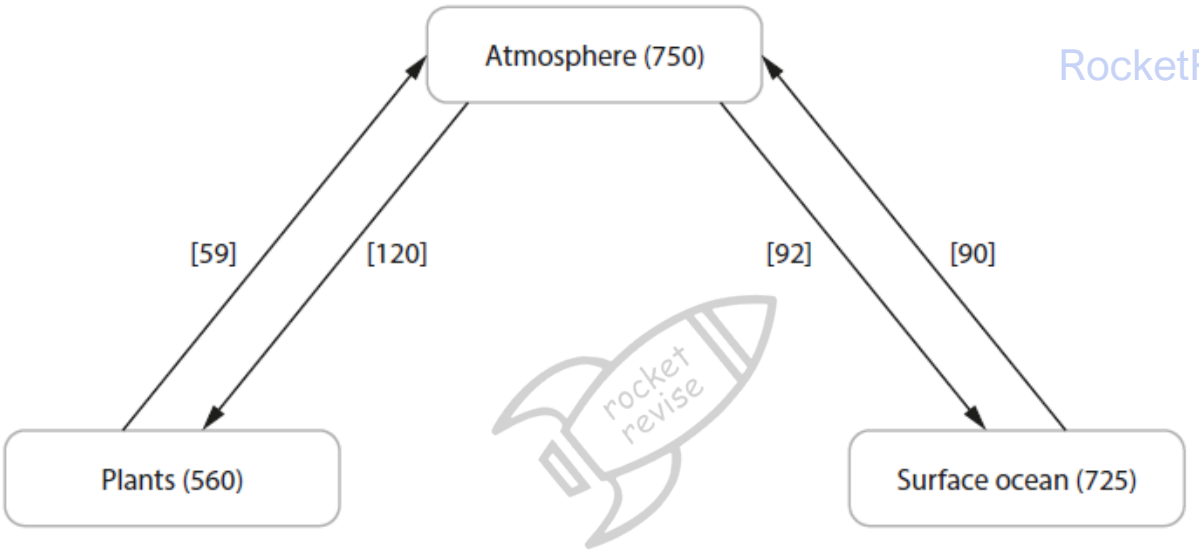
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(Total for question = 13 marks)

Q6.

The diagram shows part of the carbon cycle.



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Key

The numbers in smooth brackets ( ) show the mass of carbon in petagrams found in either the atmosphere, plants or surface ocean.

The numbers in square brackets [ ] show the mass of carbon in petagrams transferred between the atmosphere and the plants or surface ocean per year.

A petagram is equal to 10^{15} grams.

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(a) Carbon is found in carbohydrates.

The table shows some carbohydrates that may be found in plants and animals.

For each carbohydrate, put **one** cross ☒ in the appropriate box, in each row, to show where these carbohydrates are produced.

(4)

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Carbohydrate	Carbohydrate produced by			
	both plants and animals	plants but **not** animals	animals but **not** plants	neither plants nor animals
Amylose	☒	☒	☒	☒
Glucose	☒	☒	☒	☒
Glycogen	☒	☒	☒	☒
Starch	☒	☒	☒	☒

(b) The mass of carbon recycling between the atmosphere and the plants, and the atmosphere and the oceans, are roughly balanced.

(i) State how carbon is exchanged between the atmosphere and the oceans.

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(1)

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(ii) Calculate the increase in the mass of carbon in plants as a result of carbon exchange between the plants and the atmosphere in one year.

Give your answer in kilograms.

(1)

Answer ..... kg

(iii) Describe how the carbon present in sugars in the plants is returned to the atmosphere in the carbon cycle.

(3)

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(c) Anthropogenic climate change is a result of an increase in the mass of carbon in the atmosphere.

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(i) State what is meant by the term **anthropogenic climate change**.

(2)

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(ii) Explain **one** way in which the effects of anthropogenic climate change can be reduced.

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(Total for question = 13 marks)

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Q7.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

The time of death of a person can be estimated in a number of ways.

(a) The time of death can be estimated using the length of insect larvae.

What is the name of the method that uses insect larvae to estimate the time of death?

(1)

☒ **A** dendrochronology

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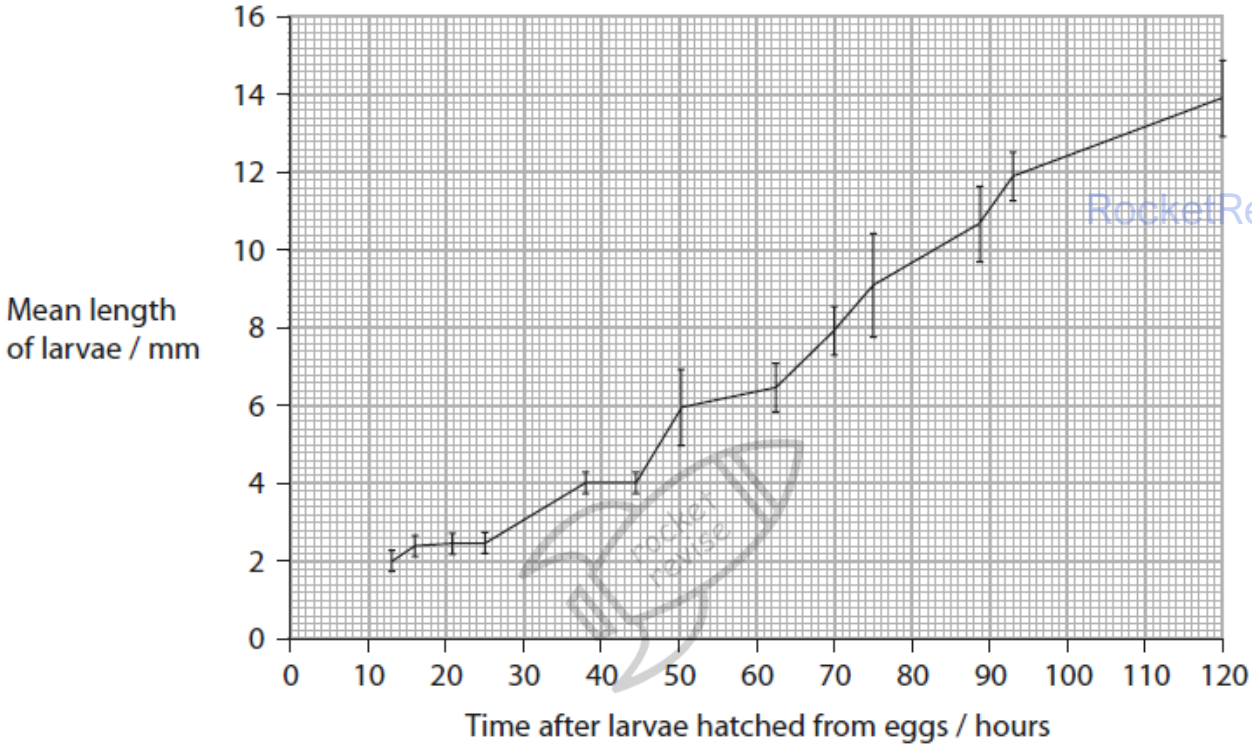
☒ **B** epigenetics

☒ **C** forensic entomology

☒ **D** species diversity

(b) The larvae of one species of blowfly can be used to estimate the time of death.

The graph shows the mean length of larvae from this species incubated at 10.62 °C.



(i) Calculate the mean growth rate of these larvae from 25 to 120 hours.

Include the units with your answer.

(2)

Answer. .....

(ii) Comment on the suitability of using this data to estimate the time of death.

Use the information in the graph to support your answer.

(3)

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(iii) Describe how the data shown in this graph could have been collected.

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(c) The time of death can be estimated using the body temperature of the corpse.

Evaluate the use of the body temperature of a corpse to estimate the time of death.

(4)

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(Total for question = 13 marks)

Q8.

Madagascar is a large island off the southeast coast of mainland Africa.

Lemurs are endemic to Madagascar.

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(a) There are many different species of lemur, all of which evolved from one common ancestor. This common ancestor is thought to be a primate that was carried across the sea from mainland Africa on a raft of vegetation.

The photographs show two different species of lemur, Sifaka and Indri, found in the same region of Madagascar.



Sifaka



Indri

The diet of Sifakas is mostly seeds but also includes fruits, flowers and some types of leaves. The diet of Indri is mainly leaves.

Explain the types of speciation that have taken place in the evolution of Sifakas and Indri in this region of Madagascar.

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(b) Lemurs have been found on another island close to Madagascar.

Scientists have used DNA profiling to show that these lemurs originated from those on Madagascar.

(i) Explain the role of the polymerase chain reaction (PCR) in DNA profiling.

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(ii) Explain how DNA profiling could show that these lemurs originated from the lemurs on Madagascar.

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(Total for question = 8 marks)

Q9.

The photograph shows a Tasmanian devil.



(Source: © Walter 64 / Alamy Stock Photo)

Tasmanian devils are found in Tasmania.

The population of Tasmanian devils has fallen drastically due to a fatal cancer.
This cancer first appeared in the mid-1990s.

There are only a few regions in Tasmania left where healthy Tasmanian devils can still be found.

(a) The number of Tasmanian devils is estimated to have fallen from 150 000 in the mid-1990s to 25 000 in 2020.

Calculate the percentage decrease in the number of Tasmanian devils from the mid-1990s to 2020.

(1)

Answer ..... %

(b) Tasmanian devils feed on the dead bodies of animals (carcasses) and scare off other scavengers.

Scientists have studied the scavenging of carcasses in two regions of Tasmania: one region where all Tasmanian devils were healthy and one region where the cancer was common.

The length of time Tasmanian devils and three other scavenging animals spent feeding on carcasses was investigated.

The results are shown in the table.

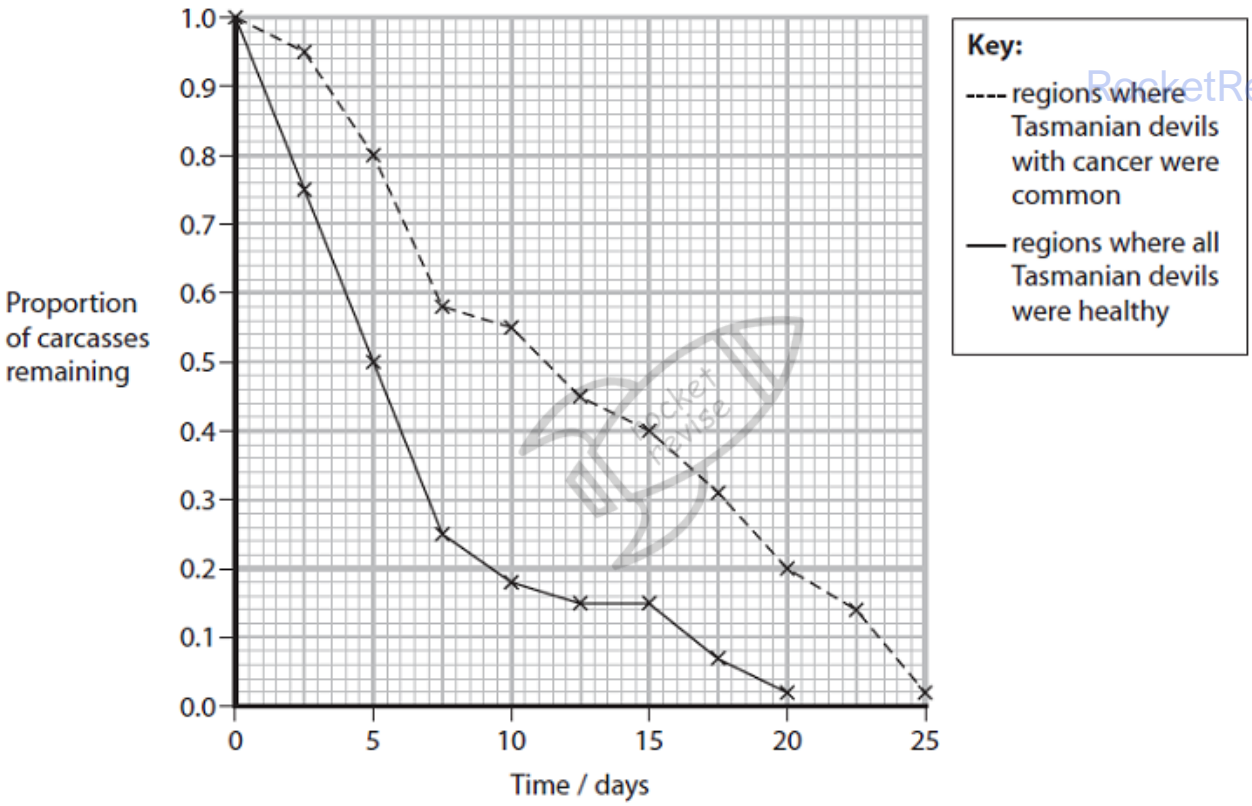
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Region	Proportion of time the scavengers spent feeding on each carcass			
	Tasmanian devil	Forest raven	Spotted-tailed quoll	Feral cat
Where all Tasmanian devils were healthy	0.76	0.13	0.06	0.05
Where Tasmanian devils with cancer were common	0.26	0.38	0.20	0.16

The length of time it took for the carcasses to be completely eaten was also investigated.

The results are shown in the graph.



*(i) Explain the results of these two studies.

Use the information in the table and the graph to support your answer.

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(ii) Explain what other **biotic** factors could have affected the results from the two different regions of Tasmania, in the second investigation.

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(Total for question = 11 marks)

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Q10.

The photographs show a Northwestern crow and an American crow.



(Source: © All Canada Photos / Alamy Stock Photo)



(Source: © MichaelGrantBirds / Alamy Stock Photo)

These two crows look very similar and are therefore difficult to distinguish as two separate species.

Scientists have studied the nuclear DNA and the mitochondrial DNA (mtDNA) of these two species of birds.

(a) Give **two** differences between the structure of nuclear DNA and mtDNA.

(2)

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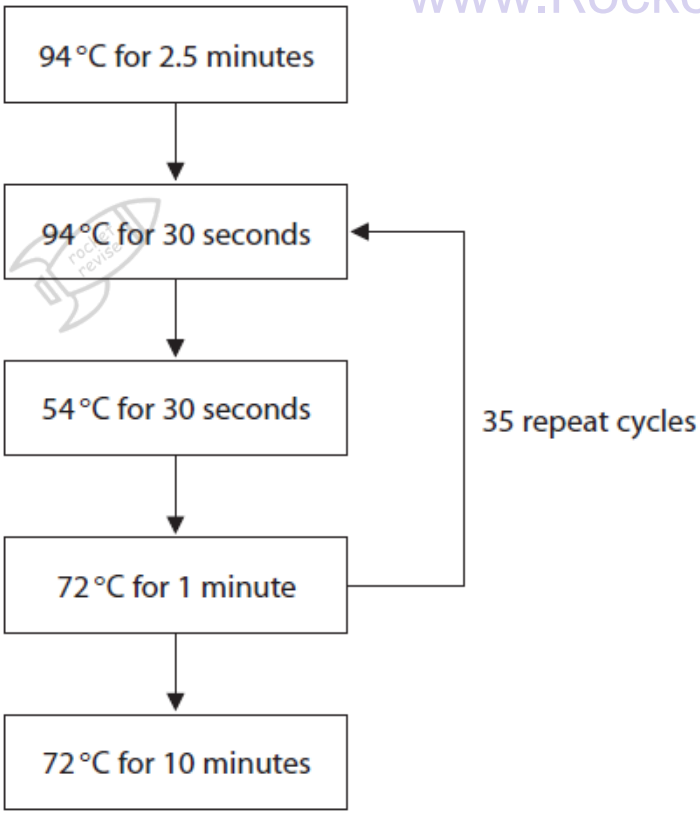
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(b) The mtDNA was isolated from these two species of crow and amplified using the polymerase chain reaction (PCR).

The diagram shows details of the process used.

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(i) Name **three** molecules, other than the mtDNA and water, that would be needed in this process.

(2)

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(ii) Calculate the total length of time, in hours, that this process would take.

Give your answer to two decimal places.

(2)

Answer ..... hours

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(iii) Explain how this process amplifies the DNA.

Use information in the diagram to support your answer.

(3)

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(c) Explain how the amplified mtDNA could be used to determine the genetic relationships between these two species of crow.

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(Total for question = 12 marks)

Q11.

Some questions must be answered with a cross in a box (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

The table shows the energy content in the three trophic levels of a food chain from an African plain.

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Trophic level	Organism	Energy content
3	Lion 	920
2	Gerenuk 	8825
1	Tree 	90480

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(a) (i) Which are the units for the energy content of these organisms?

(1)

☒ **A** kJ m^{-1} $year^{-1}$

☒ **B** kJ m^{-1} $year^{-2}$

☒ **C** kJ m^{-2} $year^{-1}$

☒ **D** kJ m^{-2} $year^{-2}$

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(ii) The energy content of the tree is the net primary productivity (NPP).

What is the relationship between NPP, gross primary productivity (GPP) and respiration (R)?

(1)

☒ **A** NPP = GPP + R

☒ **B** NPP = GPP – R

☒ **C** NPP = GPP × R

☒ **D** NPP = GPP ÷ R

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(iii) Calculate the ratio of the energy contents shown in these trophic levels:

trophic level 1 : trophic level 2 : trophic level 3

(1)

Answer .....

(iv) Explain why there is no fourth trophic level in this food chain.

(2)

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(v) Describe the role of microorganisms in recycling carbon present in this food chain.

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*(b) The photographs show gerenuk and a springbok to the same magnification.

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gerenuk



(Source: © David Havel / Alamy Stock Photo)

springbok

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The table gives some information about gerenuk and springbok.

Information	Gerenuk	Springbok
Diet	leaves from a range of different trees, often spiny trees, up to 2 m from the ground	short grass and bushes
Water requirements	met by water content of food	will drink water when available
Height / cm	80 to 105	69 to 87
Mass / kg	male 45 female 30	male 41 female 37
Colour of fur	light brown with a white underbelly	light to mid brown with large areas of white
Length of horns / cm	25 to 44 in males only	35 to 49 in both males and females but narrower horns in females

Gerenuk and springbok evolved from a common ancestor.

Deduce how the gerenuk evolved to occupy a different niche from a springbok.

Use the information in the table and the photographs to support your answer.

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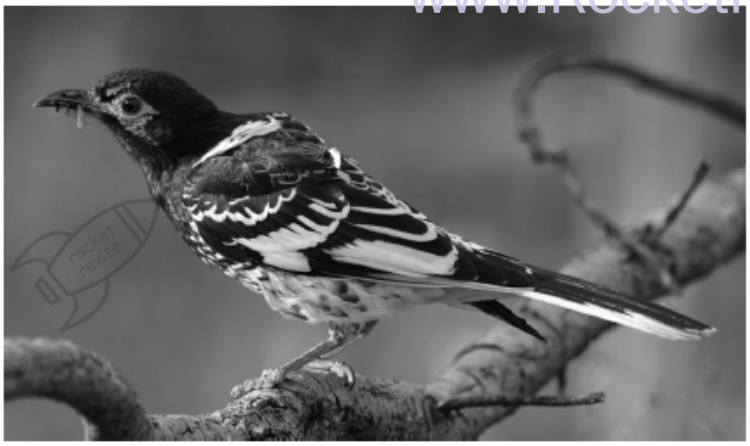
Q12.

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The photograph shows a regent honeyeater.

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(Source: © Dave Watts / Alamy Stock Photo)

(a) Regent honeyeaters are critically endangered, with fewer than 300 birds left in the wild.

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A study found that a large proportion of the male birds no longer sing the characteristic songs of their own species.

Some of the birds had learnt the songs of other species.

(i) Suggest why a large proportion of these male birds no longer sing the characteristic songs of regent honeyeaters.

(1)

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(ii) Explain the effect that this could have on the future of this species.

(3)

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(iii) Suggest how zoos could increase the number of birds in the wild that can sing the songs characteristic of this species.

(2)

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(b) Genetic studies indicate that the regent honeyeater is related to wattlebirds.

Describe how genetic studies could show this.

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(Total for question = 10 marks)

Q13.

Human serum albumin is a globular protein found in blood plasma.

Albumin has a molecular mass of 66 300 daltons.

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It consists of 585 amino acids.

At pH 7.4 albumin has over 200 negative charges on the surface of each molecule.

(a) (i) Calculate the mean molecular mass of an amino acid in albumin.

Give your answer to an appropriate number of significant figures.

(1)

Answer ..... daltons

(ii) Explain why albumin is soluble in blood plasma.

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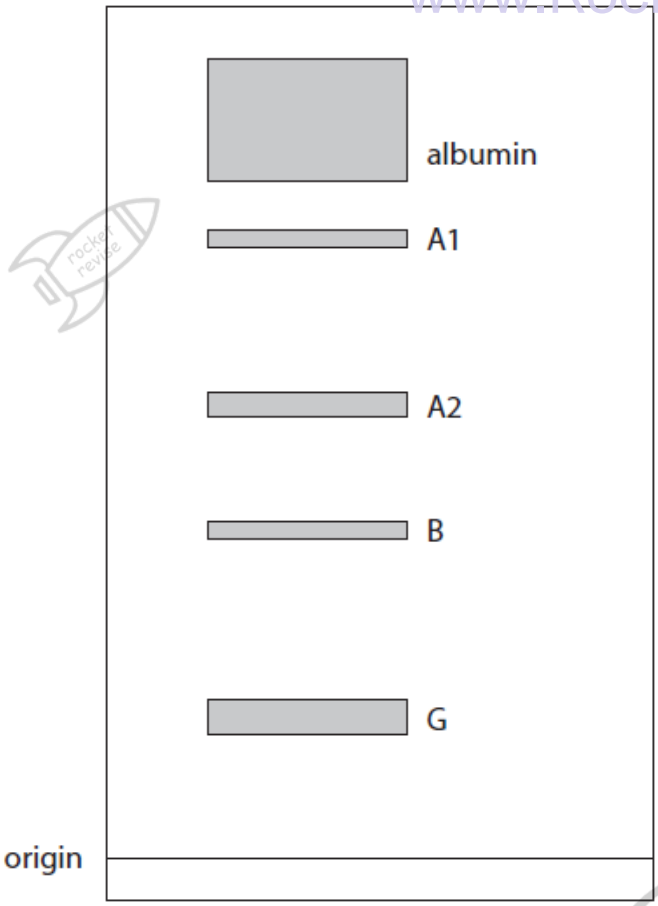
(b) Albumin can be separated from other proteins in blood plasma, A1, A2, B and G, by gel electrophoresis.

The diagram shows the banding pattern produced on separation of these proteins by gel electrophoresis.

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(i) Protein G has a concentration in blood plasma in the range 0.700 g dm^{-3} to 1.700 g dm^{-3}.

Albumin has a concentration in blood plasma in the range 0.525 g dm^{-3} to 1.275 g dm^{-3}.

Calculate the maximum difference in the concentration of these two proteins.

(1)

Answer ..... g dm^{-3}

(ii) Describe the conclusions that can be made about albumin, compared with the other four proteins present in blood plasma.

Use the information in the diagram to support your answer.

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(c) Albumin binds to and activates a proton channel, hHv1.

This proton channel is present in the membranes of sperm cells. When the channel is activated, sperm cells have the ability to penetrate and fertilise egg cells.

Sperm cells are released in a liquid called seminal fluid. Seminal fluid has an albumin concentration of 15 µmol dm^{-3}.

The fluid inside the uterus of the female has an albumin concentration of 500 µmol dm^{-3}.

Explain the role of albumin concentration in controlling fertilisation.

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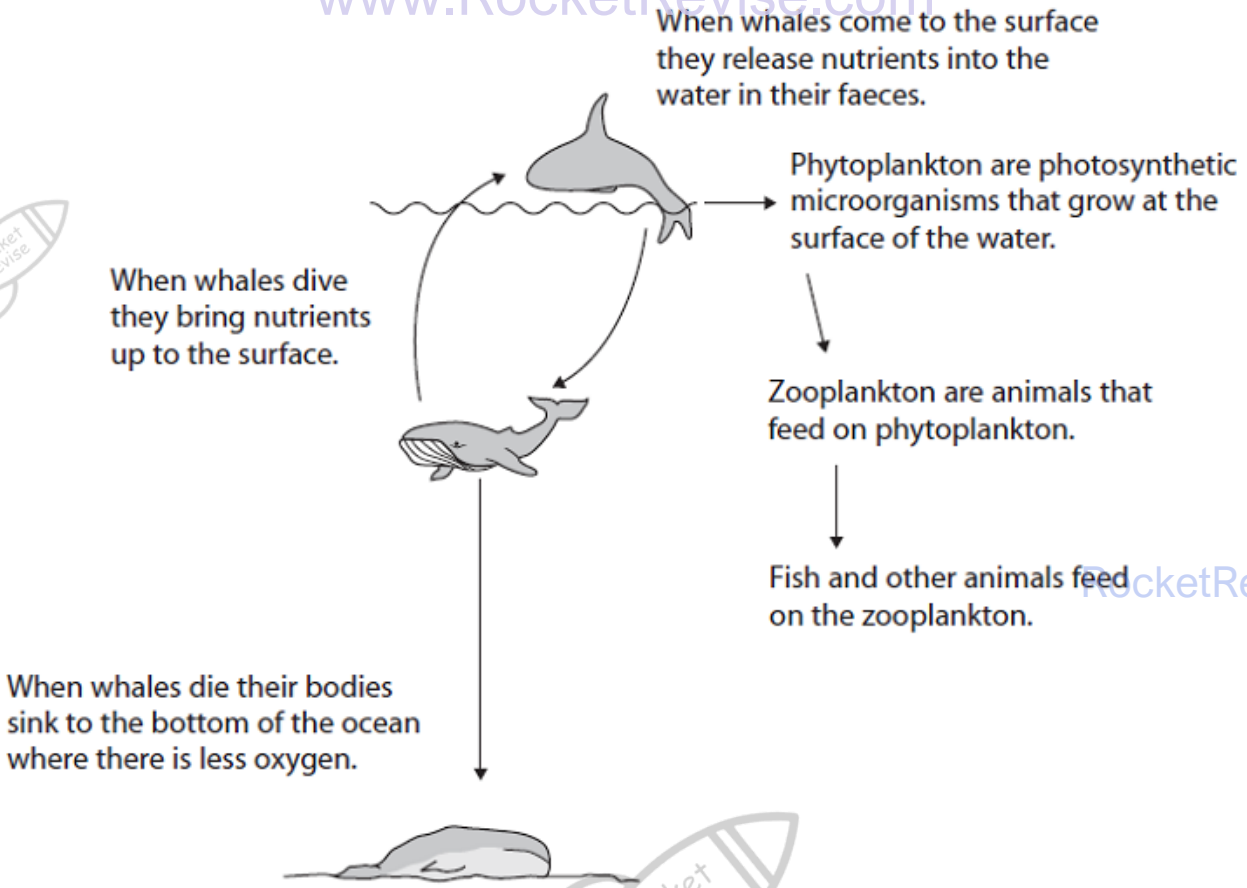
Q14.

Commercial whaling has resulted in a large decrease in the number of whales.

The diagram shows the relationships between whales and some other organisms in the ocean.

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(a) Whale faeces add nutrients containing nitrogen to the water.

(i) There are 345 Right whales in the Gulf of Maine and together they release 15.9 kg of nitrogen per day.

Calculate the mass of nitrogen released by one Right whale in a year.

(1)

Answer ..... kg per year

(ii) Nitrates contain nitrogen.

Describe the importance of nitrates to plants.

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(b) There are 1.3 million whales in the oceans.

Scientists believe that if the number of whales could be restored to their original numbers, this would have a significant effect on climate change.

(i) There were approximately 5 million whales before modern commercial whaling.

Calculate the percentage decrease in the number of whales due to commercial whaling.

(1)

Answer ..... %

(ii) Calculations have shown that if numbers could be restored, 1.7×10^9 US tons of carbon dioxide could be removed from the atmosphere per year.

A US ton is equivalent to 907 kg.

Calculate the mass of carbon dioxide, in kg, that could be removed from the atmosphere in one year.

Give your answer in standard form.

(1)

Answer ..... kg

*(iii) Discuss the possible effects of increasing the number of whales on climate change.

Use the information in the diagram and the data given in this question to support your answer.

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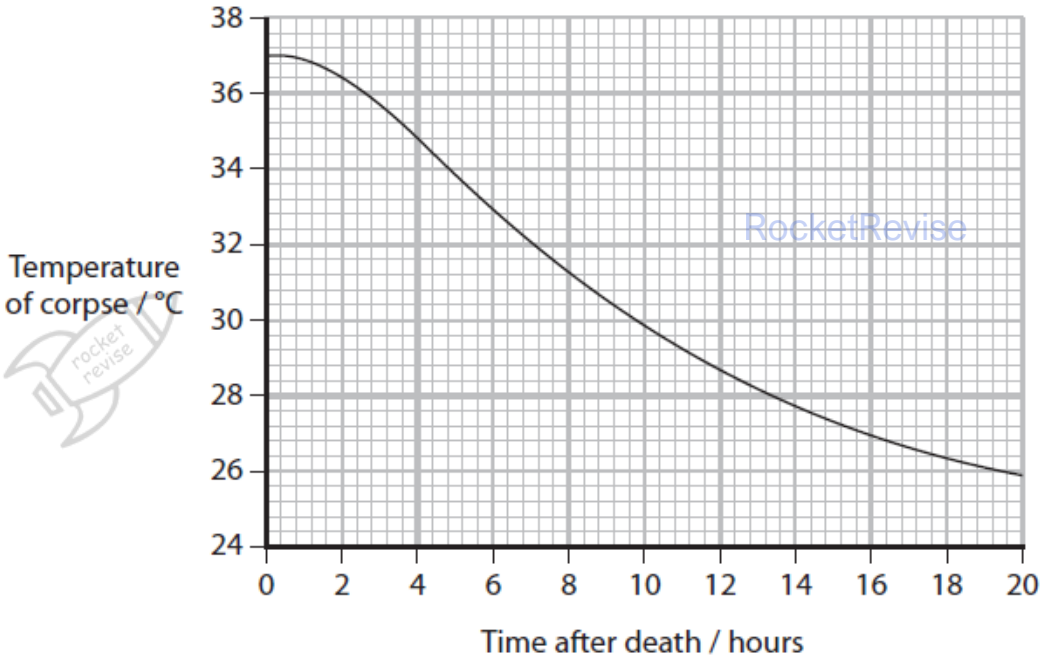
(Total for question = 11 marks)

Q15.

The time of death of a mammal can be estimated in a number of ways.

(a) Measuring the body temperature of a corpse is one method used to estimate the time of death.

The graph shows a calibration curve that can be used in this method.



(i) Describe how the time of death of a person can be estimated using this method.

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(ii) Explain the limitations of this method.

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(b) Scientists are looking for other methods to estimate the time of death, using the presence of chemicals in the blood.

One study used blood taken from five rat corpses and two pig corpses at intervals after death.

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(i) Suggest why this study used rat and pig corpses instead of human corpses.

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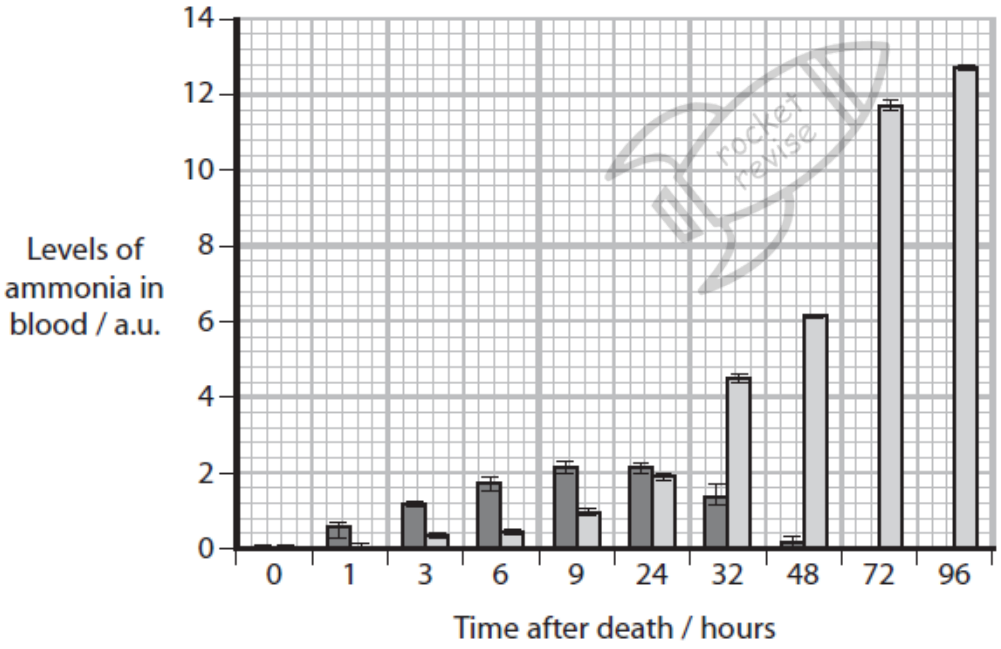
*(ii) The graphs show some of the results of this study.

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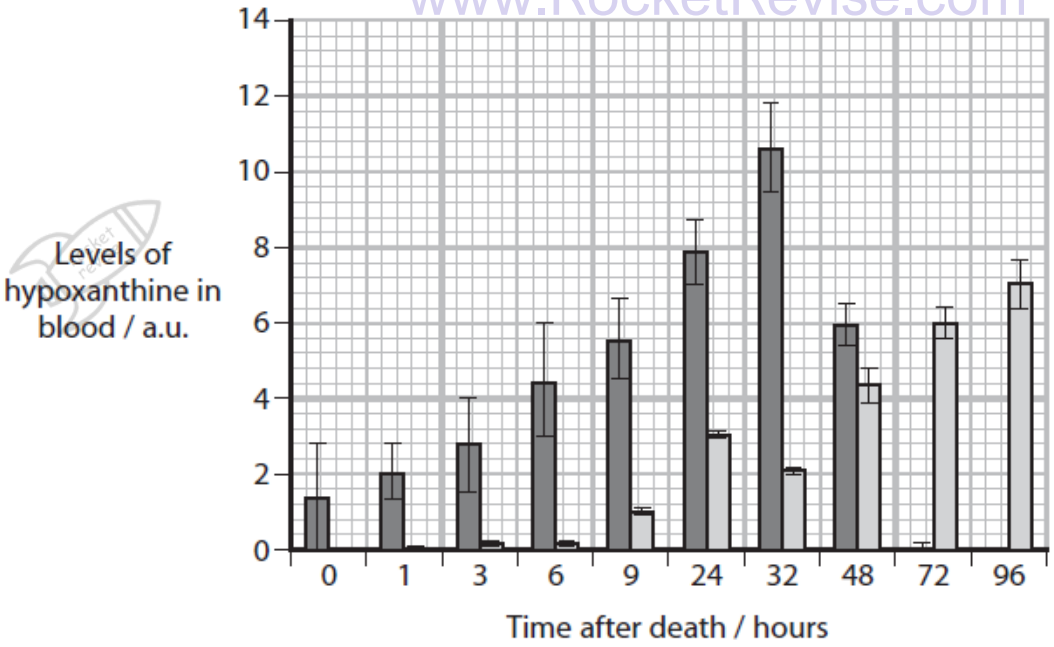
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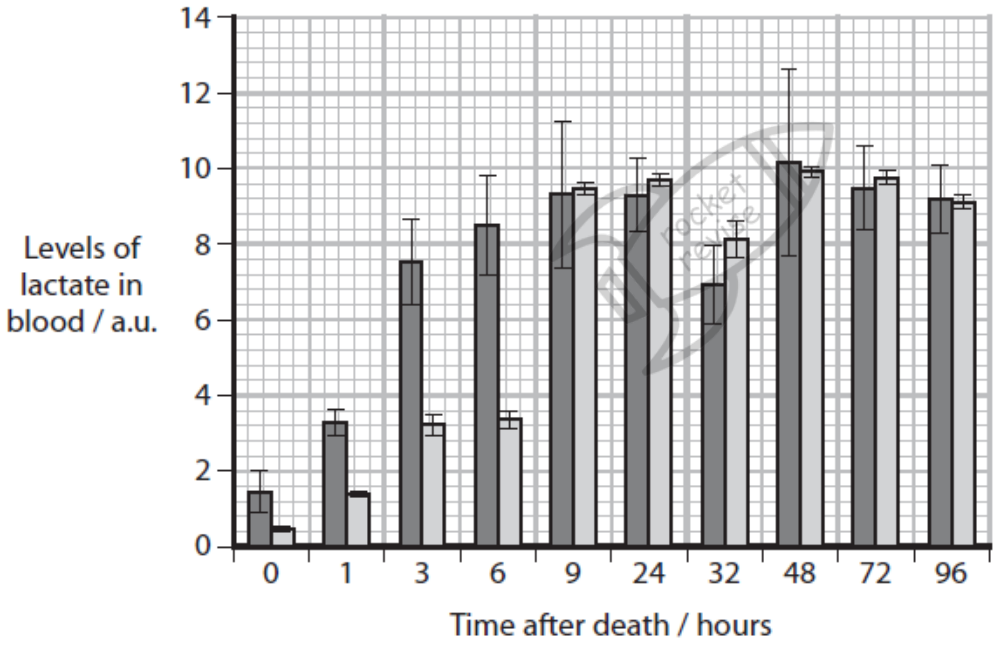
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Comment on the usefulness of using these chemicals for determining the time of death of rats and pigs.

Use the information in the graphs to support your answer.

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(iii) Discuss whether these chemical tests are likely to be useful in determining the time of death of a human.

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(Total for question = 15 marks)

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Q16.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows a black bear.



(Source: © Cindy Hopkins/Alamy Stock Photo)

Most black bears have a diet consisting of ants and berries, and hibernate during the winter months.

One very large male black bear, called Hank, was suspected of raiding nearly 40 homes in the Lake Tahoe region of America for food. Some of these raids took place during the winter months.

(a) Hank weighed 227 kg, which is 1.7 times more than the mean mass of a typical large black bear and 5 times more than the mass of a typical small black bear.

(i) Calculate the difference in mass between a typical large black bear and a typical small black bear.

(2)

Answer ..... kg

(ii) Explain why Hank was much larger than typical black bears.

(2)

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(b) Samples of DNA were collected from the homes that Hank was suspected of raiding.

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Analysis of the DNA showed that some of the raids were by two unrelated, female black bears.

(i) Which two procedures could have been used to analyse the DNA?

(1)

- ☒ **A** electron microscopy followed by PCR
- ☒ **B** entomology followed by electron microscopy
- ☒ **C** gel electrophoresis followed by entomology
- ☒ **D** PCR followed by gel electrophoresis

(ii) Describe how analysis of the DNA would show that these two other bears were female, and not related to each other or to Hank.

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(Total for question = 8 marks)

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Q17.

The photograph shows a sea turtle.

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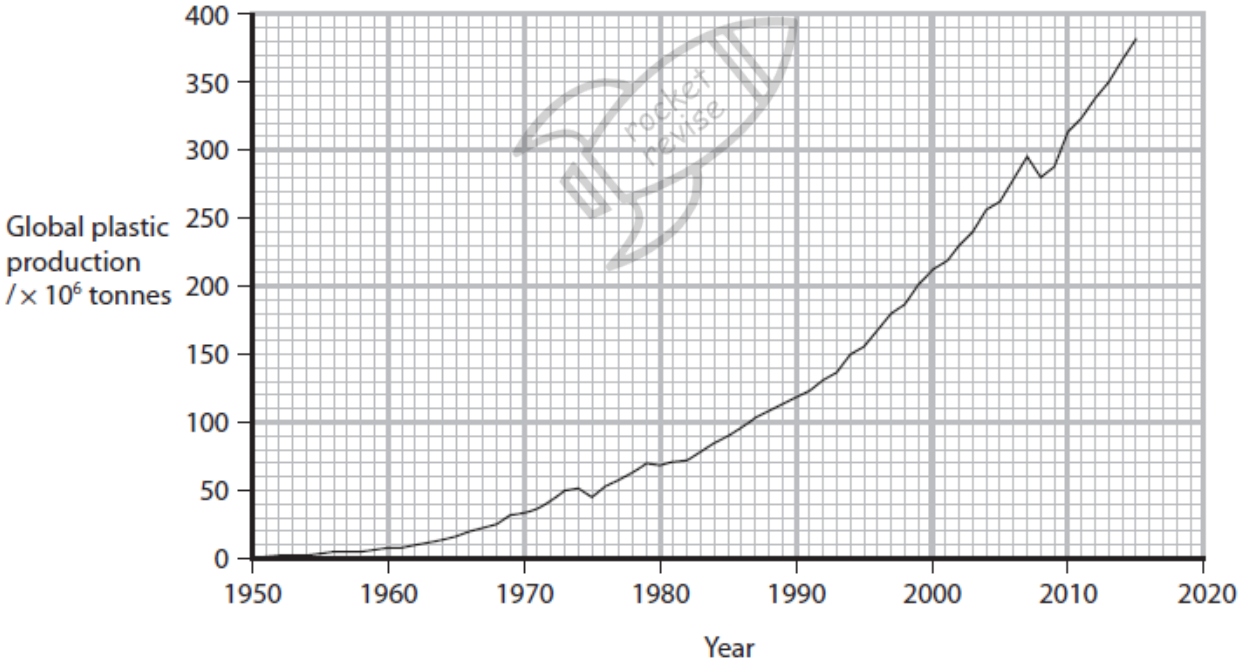


(Source: © M Swiet Productions/Getty Images)

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Sea turtles can take in plastic debris when they eat. This plastic can harm the turtles.

(a) The graph shows the global production of plastics per year from 1950 to 2015.



(i) Calculate the mean rate of global plastic production from 2000 to 2015.

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Give your answer to 3 significant figures.

(2)

Answer ..... million tonnes per year

(ii) Estimate the global plastic production in 2023.

Use your calculated value from part (i).

(1)

Answer ..... million tonnes

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(iii) Suggest why this value may be an underestimate.

Use the information in the graph to support your answer.

(1)

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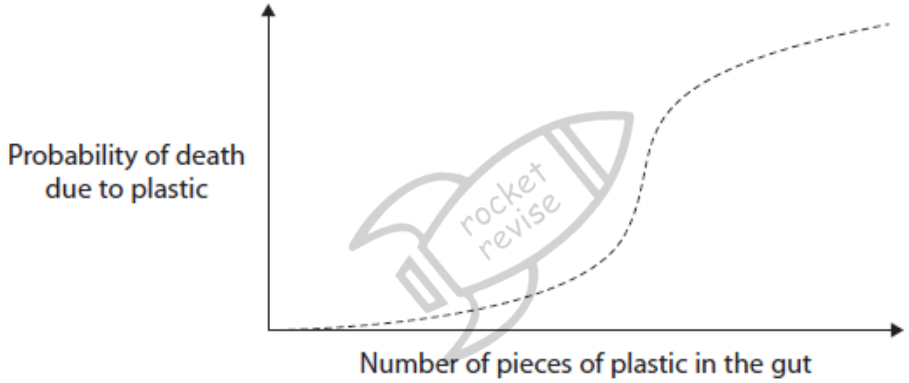
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(b) One group of scientists determined the number of pieces of plastic in the guts of individual sea turtles.

They then modelled this data to estimate the probability of death due to plastic.

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The graph shows the results of this modelling.



(i) Describe how the probability of death due to plastic increases with the number of pieces of plastic in the gut.

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(ii) Describe **two** limitations of using this graph to determine the probability of death due to the ingestion of plastic.

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*(c) Mangoes are tropical fruits that are harvested before they are ripe.

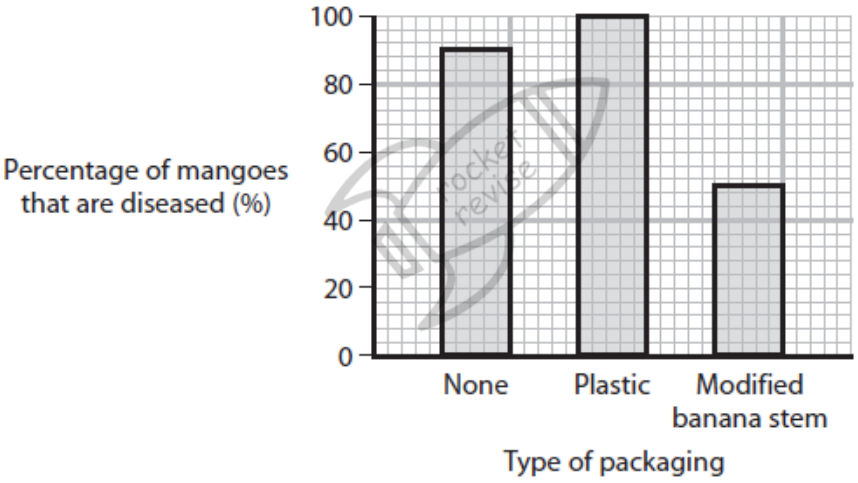
Plastic packaging is used to wrap mangoes after they have been picked. This preserves the fruit during transportation and storage.

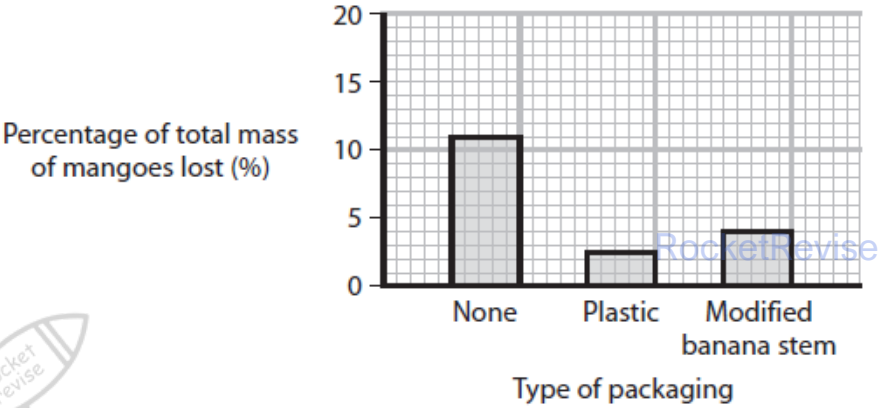
Banana stalks have a high content of cellulose and have been modified to use as packaging material for mangoes.

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A study compared the use of these two types of material and their decomposition.

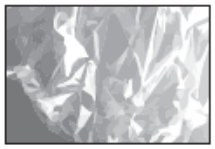
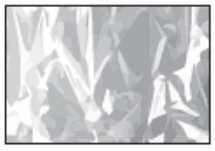
The graphs and the table show the results of this study.





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Time of decomposition of packaging / weeks	Appearance of packaging	
	Plastic packaging	Modified banana packaging
0		
2		
4		

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Discuss the advantages of using packaging made from modified banana stalks.

Use the information in the graphs, the table and your own knowledge to support your answer.

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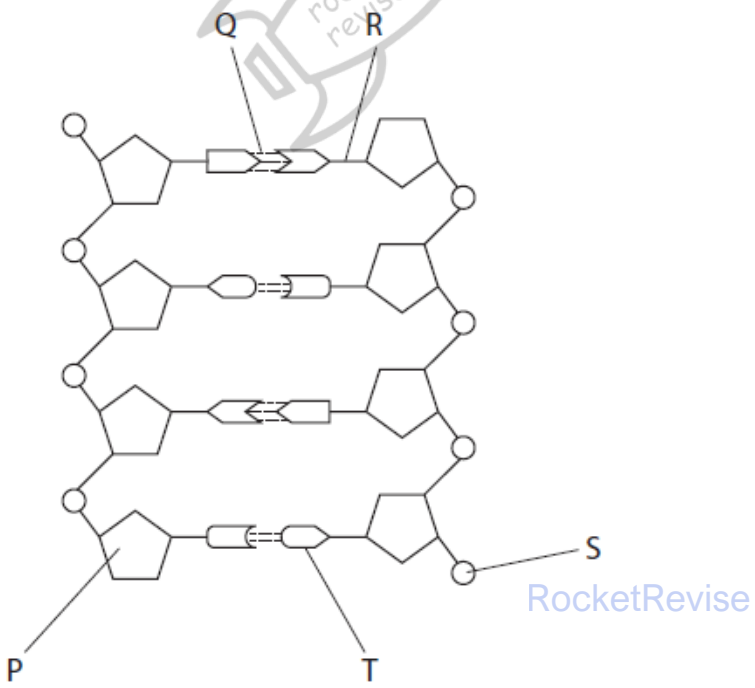
Q18.

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Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Analysis of DNA samples taken from endangered species can be used in the conservation and management of these species.

(a) The diagram shows part of a molecule of DNA.



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(i) Which row of the table identifies the bonds **Q** and **R**?

(1)

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	Bond Q	Bond R
☒ **A**	covalent	hydrogen
☒ **B**	hydrogen	covalent
☒ **C**	hydrogen	phosphodiester
☒ **D**	phosphodiester	covalent

(ii) Which is a negatively charged part of a DNA molecule?

(1)

☒ **A** P

☒ **B** R

☒ **C** S

☒ **D** T

(b) The Maleo is an endangered bird endemic to Indonesia.

The photograph shows a Maleo, digging a hole in sand to lay her egg.



(Source: © Avalon.red/Alamy Stock Photo)

Maleos live in the forest and only come to the beaches to lay their eggs. Once the eggs are laid and buried in the sand, the birds return to the forest.

When the egg hatches the young bird digs its way out of the sand and can fly straightaway.

Habitat fragmentation is thought to be partly responsible for this species being endangered. The forests are separated from the beaches by the activities of humans.

(i) State the meaning of the term **endemic**, as used in the context of this question.

(1)

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(ii) Give **one** reason why habitat fragmentation could endanger these birds.

(1)

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(iii) Suggest the advantages of:

- laying the egg in the sand
- the adults leaving the site
- the young birds being able to fly.

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(iv) In a genetic study of these birds, DNA from the developing chick was isolated from the remains of the eggshell and amplified using PCR.

To amplify a DNA molecule using PCR, two different primers are needed.

Explain why two different primers are needed to amplify one molecule of DNA in a PCR.

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(v) These birds are thought to be monogamous, which means that they only mate with one bird of the opposite sex.

Birds are only monogamous if the ratio of male to female birds is 1 : 1.

In one genetic study, DNA from 23 eggshells was analysed.

This study found that 8 of the chicks were female and 15 chicks were male.

Comment on the claim that Maleos are monogamous.

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(Total for question = 12 marks)

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Q19.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The photograph shows part of a vanilla plant, grown in tropical areas along the Indian Ocean.

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(Source: © Babelon Pierre-Yves/Alamy Stock Photo)

Vanilla is an expensive spice to produce due to labour costs and damage to crops by pests.

(a) Viruses such as the vanilla mosaic virus damage the leaves of vanilla plants.

The vanilla mosaic virus has a similar structure to tobacco mosaic virus (TMV).

Which describes TMV?

(1)

☒ **A** DNA virus with an envelope

☒ **B** DNA virus without an envelope

☒ **C** RNA virus with an envelope

☒ **D** RNA virus without an envelope

(b) Disposing of plastic waste is a global problem as it is very difficult to recycle.

(i) In 2018, 8.3×10^{12} kg of plastic was produced.

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Only 8.7 % of this plastic was recycled.

Calculate the mass of plastic that was **not** recycled.

Express your answer in standard form to one decimal place.

(2)

Answer ..... kg

(ii) Scientists are developing methods that use microorganisms to break down plastics.

Some of the products produced are being used for other purposes, such as vanilla flavouring.

Explain why some microorganisms can break down plastics.

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(iii) One type of bacteria, *Pseudomonas*, uses half of the plastic it breaks down to produce its own biomass, with the rest released as carbon dioxide.

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Explain how the breakdown products become biomass and carbon dioxide.

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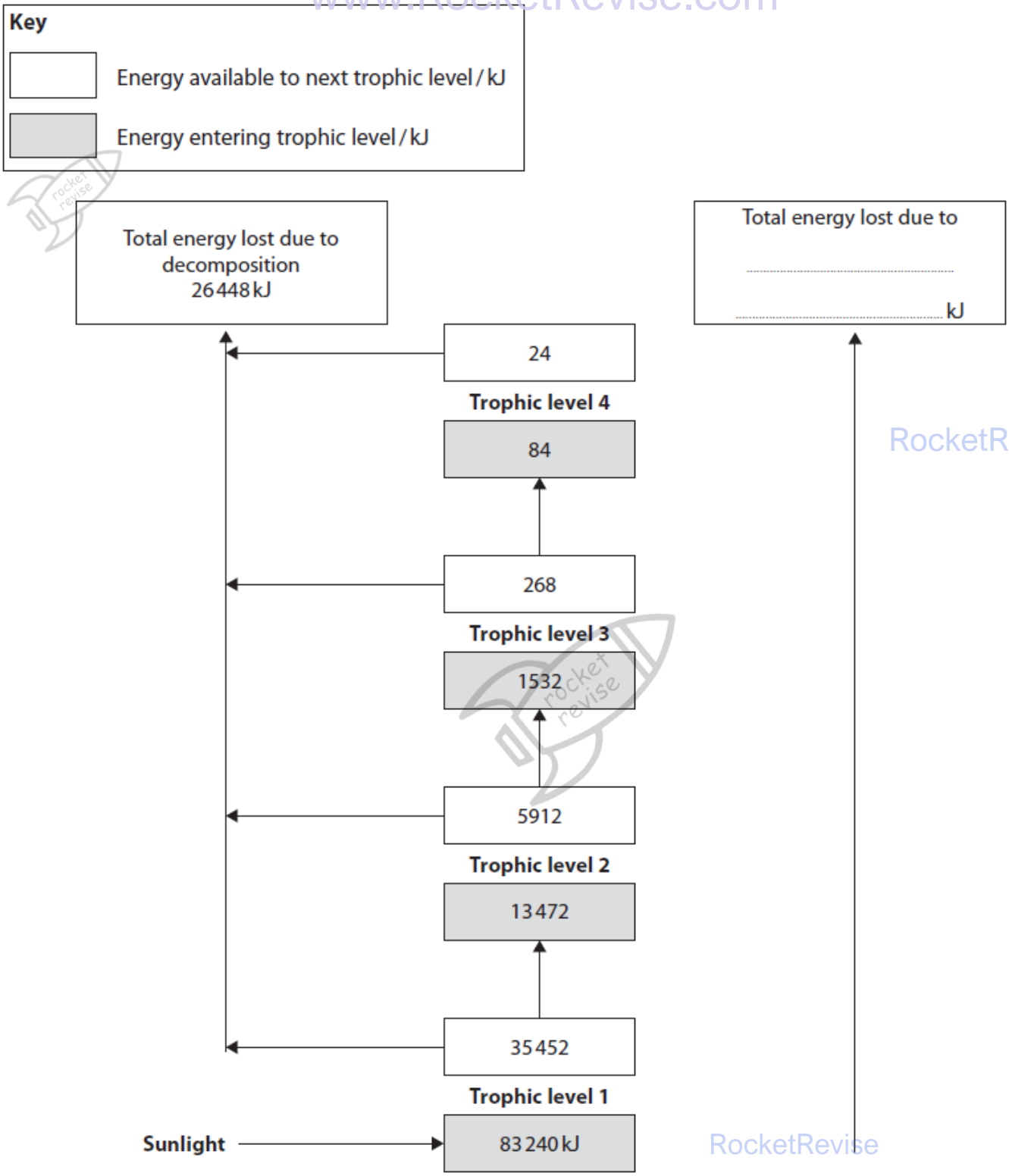
Q20.

The trophic level of an organism is the position it occupies in a food chain.

(a) The diagram shows the energy flow between trophic levels of a food chain.

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(i) Calculate the net primary productivity (NPP) as a percentage of the gross primary productivity (GPP) in this food chain.

(2)

Answer ..... %

(ii) Complete the right-hand side of the diagram by adding:

- arrows
- the missing information in the box.

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(iii) Explain the importance of enzymes produced by microorganisms in decomposition.

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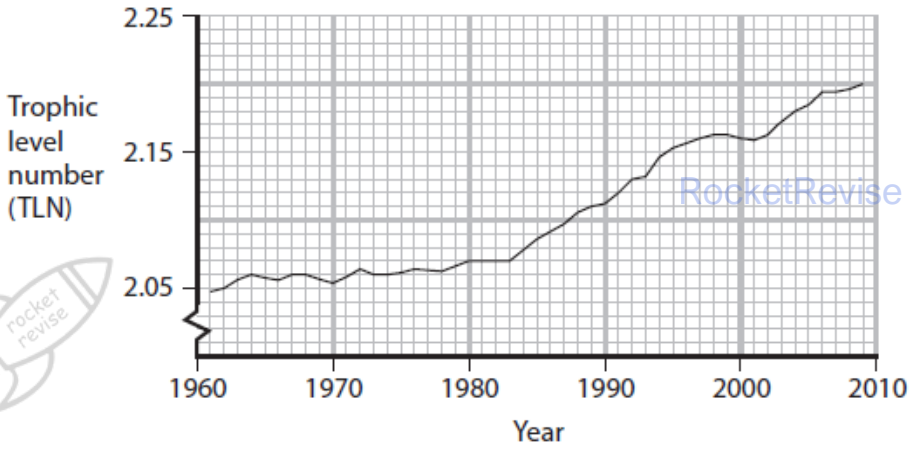
(b) Some animals eat a diet of meat and plant material.

These animals can be given a trophic level number (TLN) based on a scale of 1 to 5.

Animals that eat only plants have a TLN of 1 and animals that eat only meat and are at the top of the food chain have a TLN of 5.

Humans are an example of animals that eat both plants and animals.

The graph shows the change in the TLN for the total population of China and India from 1960 to 2010.



Explain what the information in the graph indicates about the diet of this population during this time period.

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Q21.

Some disorders are caused by mutations in the mitochondrial DNA and can be inherited.

Three-parent embryos have been developed to prevent the inheritance of these disorders.

Diagram 1 shows how three-parent embryos are made.

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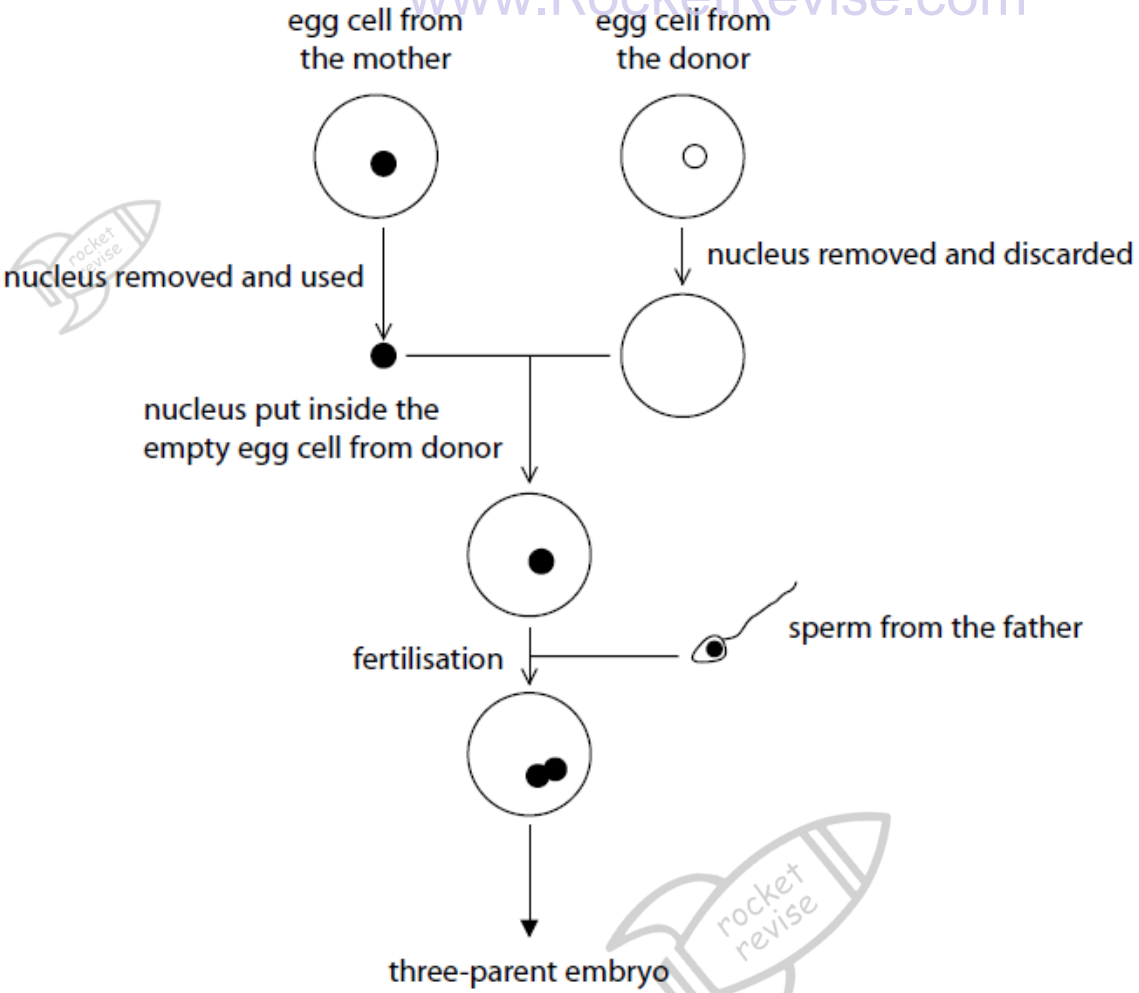


Diagram 1

(a) The cells in the three-parent embryo are totipotent stem cells.

State what is meant by the term **totipotent stem cells**.

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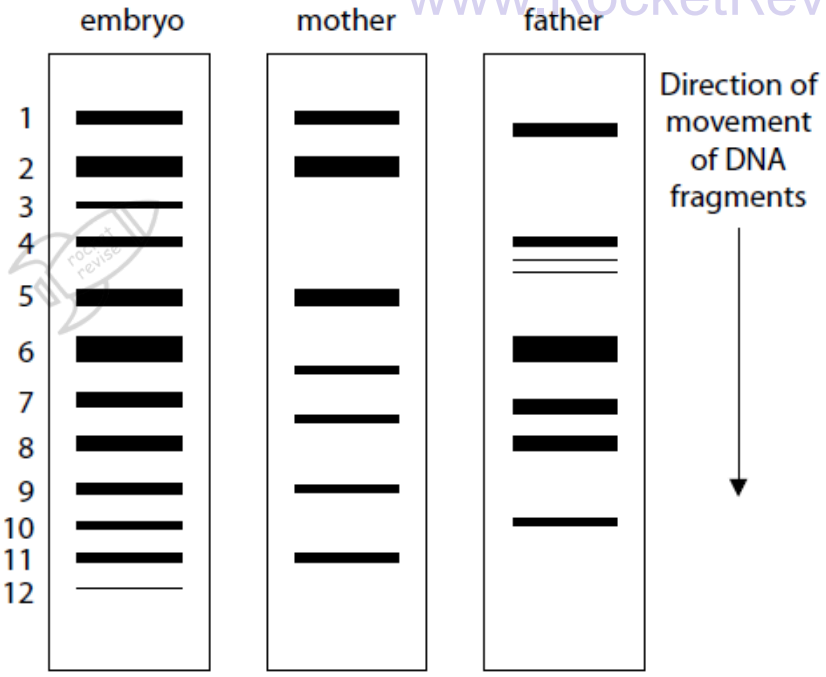
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(b) Diagram 2 shows the results of separating the DNA from the three-parent embryo, the mother and the father, using gel electrophoresis.

The bands from the DNA of the embryo are labelled from 1 to 12.

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Diagram 2

(i) Which of the following is the reason for the movement of the DNA fragments?

(1)

☒ **A** negatively-charged fragments move towards the negative end of the gel

☒ **B** negatively-charged fragments move towards the positive end of the gel

☒ **C** positively-charged fragments move towards the negative end of the gel

☒ **D** positively-charged fragments move towards the positive end of the gel

(ii) Which band on the gel contains the heaviest DNA fragments?

(1)

☒ **A** 1

☒ **B** 3

☒ **C** 6

☒ **D** 9

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*(iii) Explain the banding pattern of this three-parent embryo. Use information in Diagram 1 and Diagram 2 to support your answer.

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Q22.

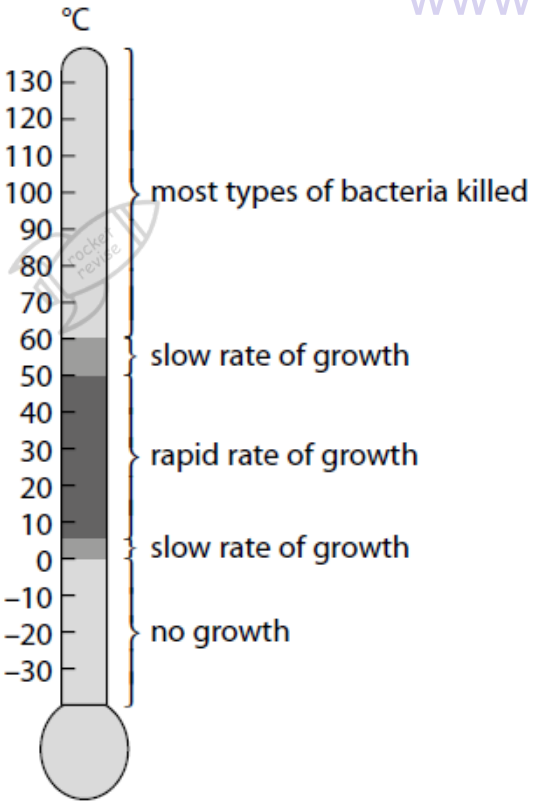
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Temperature affects the rate of growth of bacteria.

The diagram shows some information about the growth of bacteria in different ranges of temperature.

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(a) Explain why most types of bacteria are killed at temperatures above 60 °C, but bacteria can grow slowly in a temperature range of 50 °C to 60 °C.

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(b) (i) The growth rate constant will be at its highest in the temperature range of 5 °C to 50 °C.

Calculate the growth rate constant (k) of bacteria that have increased from 5×10^3 cells per cm^3 to 1.3×10^5 cells per cm^3 in 4 hours.

(3)

$$k = \frac{\log_{10} N_t - \log_{10} N_0}{0.301 \times t}$$

Answer .....

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(ii) The formula used to calculate the growth rate constant can only be applied to one phase of bacterial growth.

To which phase of bacterial growth can the formula be applied?

(1)

☒ **A** death

☒ **B** exponential

☒ **C** lag

☒ **D** stationary

(c) (i) Explain why some foods are kept in refrigerators at a temperature between 0 °C and 5 °C.

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(ii) Explain why there is no growth of bacteria in a freezer at a temperature of −18 °C.

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(Total for question = 13 marks)

Q23.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Broccoli is a vegetable plant that is grown as a crop in many parts of the world.

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Broccoli is susceptible to a disease called clubroot.

Genetic engineering can be used to produce clubroot-resistant varieties of broccoli.

Genetic engineering involves inserting resistance genes into the cells of the broccoli plants.

The resistance genes were obtained from linseed plants.

Scientists can use DNA profiling to identify the broccoli plants that have been successfully genetically engineered.

(a) Samples of DNA were extracted from the plants and subjected to a polymerase chain reaction (PCR) before analysis.

(i) Explain why PCR was used on the DNA samples.

(2)

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(ii) Taq (DNA) polymerase is used in PCR.

Describe the role of Taq (DNA) polymerase.

(2)

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(b) After PCR, gel electrophoresis was used to separate the DNA fragments.

(i) Which type of enzyme is used to produce DNA fragments?

(1)

☒ **A** integrase

☒ **B** restriction enzyme

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☒ **C** reverse transcriptase

☒ **D** RUBISCO

(ii) The results of the gel electrophoresis were used to identify the plants that contained the resistance genes.

Deduce the types of plants that would have supplied the DNA samples for this analysis.

Give reasons for your answer.

Use the information in the question to support your answer.

(3)

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(Total for question = 8 marks)

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Q24.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Leaves that have fallen from trees are an important source of nutrients for many organisms living in streams.

Bacteria and fungi can break down molecules such as cellulose and lignin, releasing nutrients for small animals that live in the water.

Decomposition of two types of leaf in the Mahoning River, Ohio, was investigated.

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(a) (i) Which row of the table describes the structure of cellulose?

(1)

	Type of glucose molecule	Type of glycosidic bond
☒ **A**	alpha (α)	1–4
☒ **B**	alpha (α)	1–6
☒ **C**	beta (β)	1–4
☒ **D**	beta (β)	1–6

(ii) Cellulose and lignin are broken down by enzymes released from the bacteria and fungi.

Fungi belong to the domain Eukarya.

Which of the following statements are correct for **both** bacteria and fungi?

1 The enzymes are synthesised on ribosomes

2 The genes coding for these enzymes are found in the nucleus

3 Enzymes are released by endocytosis

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(1)

☒ **A** 1 only

☒ **B** 1 and 2 only

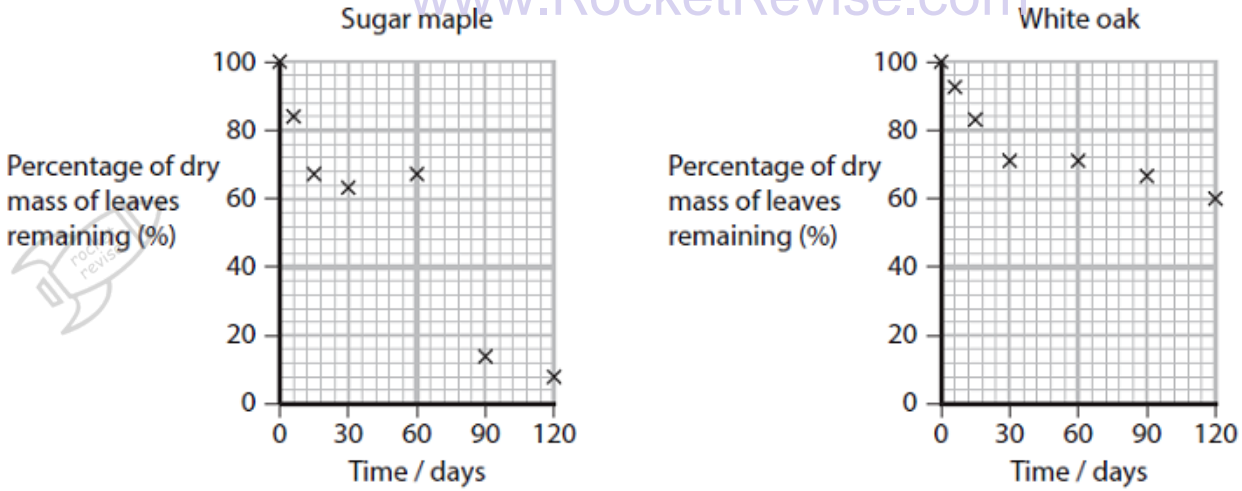
☒ **C** 2 only

☒ **D** 2 and 3 only

(b) The graphs show the extent to which two types of leaf decomposed in 120 days.

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(i) Decomposition was measured as a decrease in dry mass of the leaves.

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Suggest why dry mass was measured and not wet mass.

(1)

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(ii) Calculate the rate of decrease in percentage dry mass between 30 days and 120 days for the sugar maple leaves.

Give your answer to **two** significant figures.

(2)

Answer .....

(iii) Describe **two** conclusions that can be made about the decomposition of sugar maple leaves compared with the white oak leaves.

(2)

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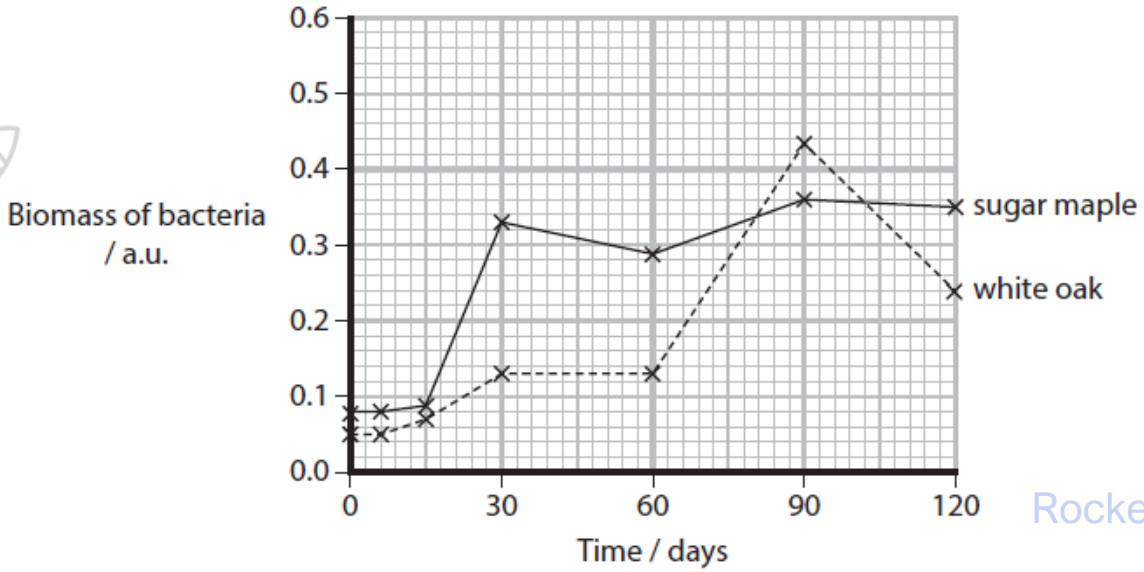
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(c) The biomass of bacteria on the two types of leaf during the 120-day period was determined.

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The graph shows the changes in biomass of bacteria on these leaves.



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(i) Calculate how many times greater the biomass of bacteria on the sugar maple leaves is after 30 days compared with the biomass of bacteria at 0 days.

(1)

Answer .....

(ii) Suggest why the increase in biomass of bacteria after 30 days is greater on the sugar maple leaves than it is on the white oak leaves.

Give a reason for your answer.

(2)

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(iii) Suggest why the biomass of bacteria on the white oak leaves decreased after 90 days.

Give a reason for your answer.

(2)

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(Total for question = 12 marks)

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Q25.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The time of death of a mammal can be determined by a number of different methods.

(a) Insects can be used to determine the time of death of a mammal.

(i) Which is the term used to describe the use of insects in determining the time and place of death of a mammal?

(1)

☒ **A** chromatography

☒ **B** dendrochronology

☒ **C** forensic entomology

☒ **D** heterozygosity

(ii) Which of the following factors provides information about the time of death of a mammal?

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1. species of insect

2. size of larvae, e.g. maggots

3. stages in development of insects

(1)

☒ **A** 1 only

☒ **B** 1 and 3 only

☒ **C** 2 and 3 only

☒ **D** 1, 2 and 3

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(b) The body temperature of a human corpse can be used to determine the time of death.

(i) Which is the most accurate method of measuring the body temperature of a human corpse to determine time of death?

(1)

☒ **A** placing a temperature-sensitive strip on the forehead

☒ **B** pushing a temperature probe into the liver

☒ **C** putting a thermometer under the tongue

☒ **D** putting a temperature probe under the armpit

(ii) Time of death can be estimated by determining the fall in body temperature.

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Which row of the table shows how more clothing and higher ambient temperature would affect the estimated time of death?

(1)

		Effect on estimated time since death of:	
		more clothing	higher ambient temperature
☒	**A**	increase	increase
☒	**B**	increase	decrease
☒	**C**	decrease	increase
☒	**D**	decrease	decrease

(c) The extent of muscle contraction (rigor mortis) can be used to estimate the time of death.

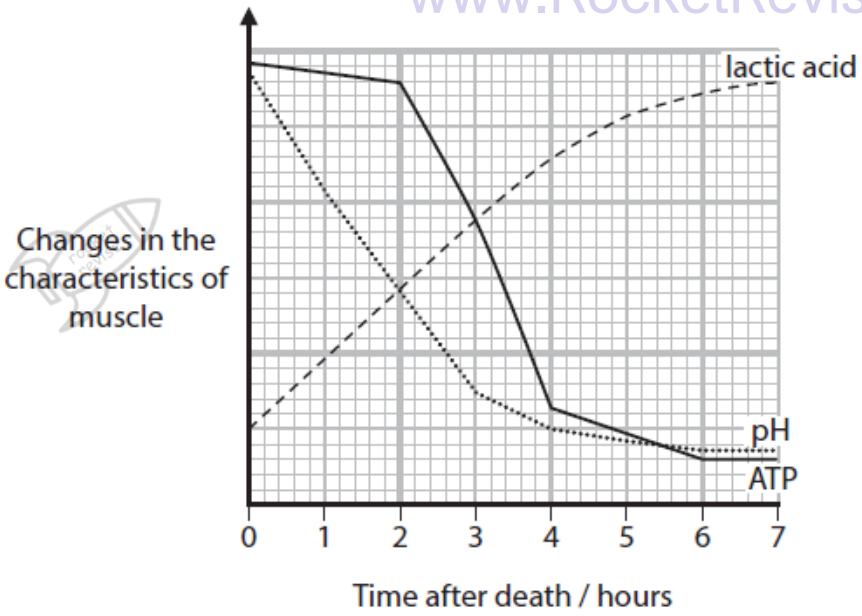
(i) The graph shows the changes in three characteristics of muscle with time after the death of a mammal:

- ATP
- lactic acid
- pH

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Explain the changes in each of the **three** characteristics shown in this graph.

Use your own knowledge to support your answer.

(3)

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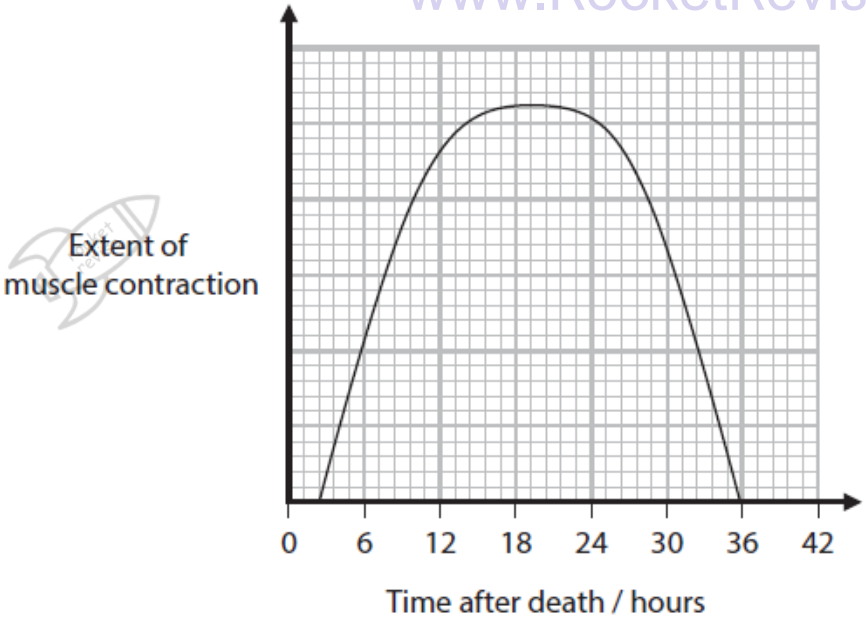
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(ii) The graph shows the changes in the extent of muscle contraction with time after death.

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Explain why using the extent of muscle contraction by itself may give an incorrect time of death.

Use the information in the graph to support your answer.

(3)

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(Total for question = 10 marks)

Q26.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

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The time of death of a mammal can be determined in a number of ways.

(a) The identification of insects present on the body can be used to estimate how long a mammal has been dead.

Which method uses the numbers and species of insects to estimate how long a mammal has been dead?

(1)

☒ **A** chromatography

☒ **B** dendrochronology

☒ **C** evolution

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☒ **D** forensic entomology

(b) Explain why body temperature and the degree of muscle contraction can be used to determine the time of death of a mammal.

(4)

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(Total for question = 5 marks)

Q27.

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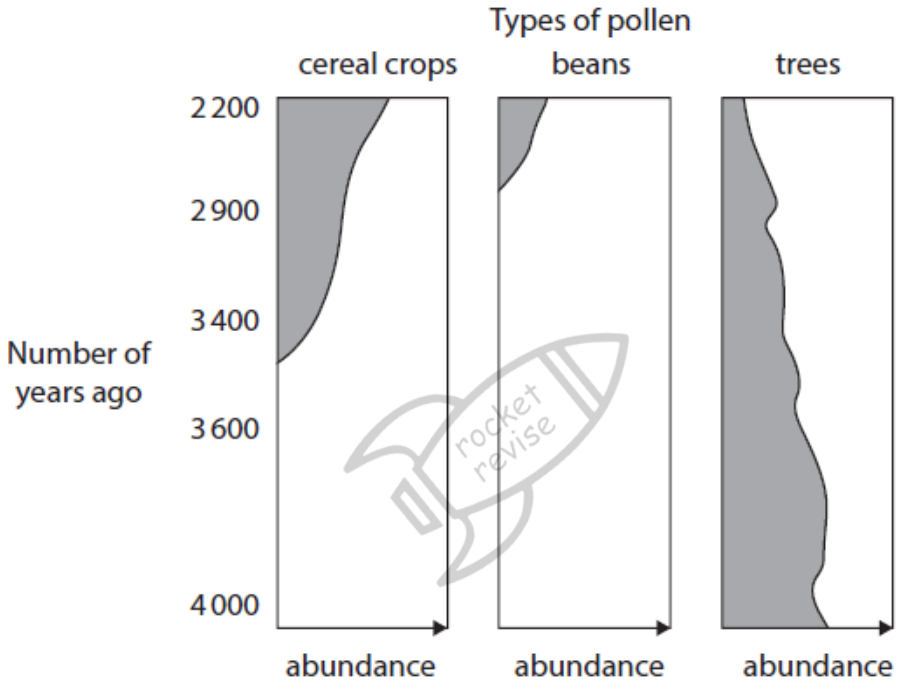
Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Peat bogs are waterlogged areas with low pH.

(a) Pollen grains in peat bogs can be used to determine the types of plants that grew in a particular area in the past.

The diagram shows the types of pollen (shaded areas) found in a site in New Mexico.

Cereal crops and beans are sources of food for humans and their animals.



Explain why the abundance of these types of pollen changes, as shown in this diagram.

(3)

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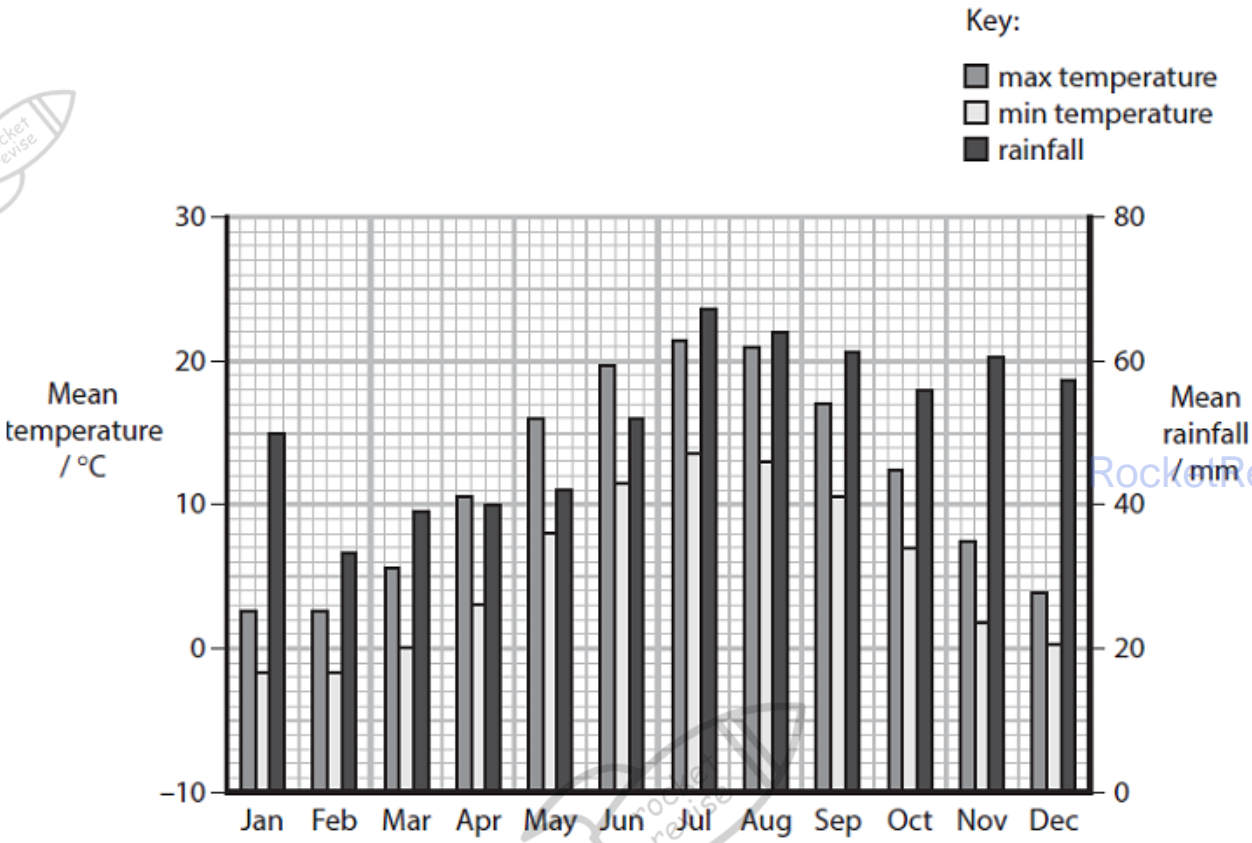
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(b) In 1950, the preserved body of a man was found in a peat bog in Denmark. The man died over 2 000 years ago.

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The graph shows the maximum and minimum mean temperature and the mean rainfall in Denmark.



(i) Which is the difference between the maximum and minimum temperatures in May?

(1)

☒ **A** 5°C

☒ **B** 8°C

☒ **C** 10°C

☒ **D** 16°C

(ii) Explain how the conditions in the peat bog preserved the body of this man.

Use the information in the question and your own knowledge to support your answer.

(5)

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(Total for question = 9 marks)

Q28.

Answer the questions with a cross in the boxes you think are correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

The time of death of a mammal can be determined by measuring the body temperature of the corpse.

(a) Which of the following can also be used to determine the time of death of a mammal?

(1)

☒ **A** dendrochronology and degree of muscle contraction

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☒ **B** degree of muscle contraction and extent of decomposition

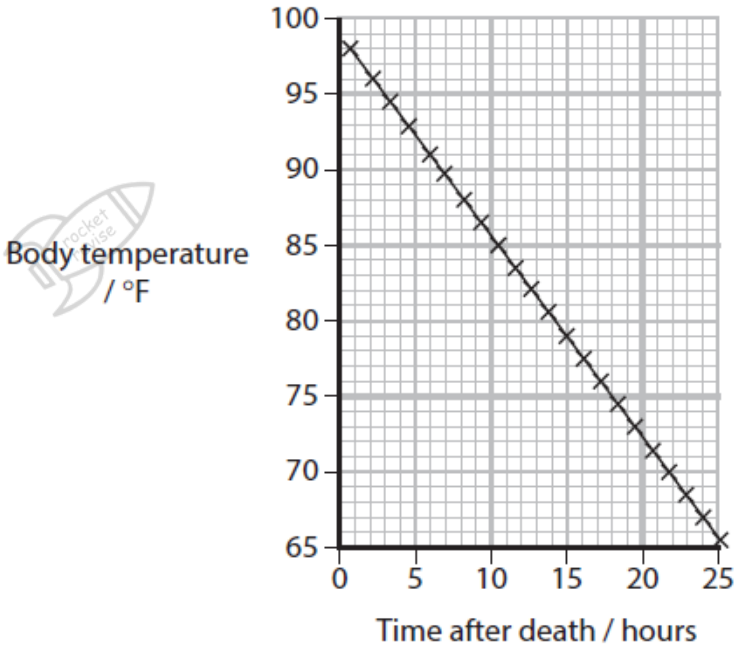
☒ **C** extent of decomposition and polymerase chain reaction

☒ **D** polymerase chain reaction and dendrochronology

(b) The graph shows a cooling curve that can be used to determine the time of death of a mammal.

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(i) A mammal had been dead for 15 hours.

Calculate the body temperature of this mammal in °C.

Use the formula:

$$\text{Temperature in °C} = \frac{5 \times (\text{temperature in °F} - 32)}{9}$$

Give your answer to one decimal place.

(2)

Answer ..... °C

(ii) Explain how this cooling curve would change if the mammal had more body fat.

(3)

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(iii) The equation for the cooling curve shown in the graph can be used to determine the time of death.

Write the equation for this cooling curve, using the format:

$$y = mx + c$$

(3)

Answer .....

(Total for question = 9 marks)

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